

Hedging the FX Carry Book with Futures

Can funding-currency, Treasury, S&P 500, DXY, or VIX futures hedge the 27-currency carry book — statically or dynamically?

arjun/hedge_comparison.ipynb
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Sample 2007-05 → 2026-06 · Bloomberg
static pull

0.466

Unhedged book — net Sharpe, 27-ccy
G10+EM carry (committed &
reconciled)

0.571

The bar to beat: **de-risk control** —
book off during FX-vol stress

+34 bp/day

JPY overlay payoff on the book's
worst-5% days — best tail hedge
tested

−1.8 %/yr

...and its cost: carry paid every day to
stay long JPY (earnable pricing)

01 WHAT THE NOTEBOOK DOES

The book. Rank 27 currencies by forward-implied carry; long high-yielders, funded by shorting the low-yielders (JPY, CHF). Vol-targeted 10%, net of costs. A reconciliation gate reproduces the committed Sharpes exactly before any test runs.

Part A — the JPY hedge, three ways. The book is short JPY ~79% of days *on purpose* (its best G10 earner), so a plain static long-JPY overlay is just un-sizing a profitable leg — not a hedge. The notebook therefore tests the versions that could work: **(A.1)** static hedge-ratio scaling of the JPY leg, **(A.2)** cutting or flipping the funding legs only when a tradeable FX-vol trigger fires, and **(A.3)** a variance-minimising JY1 overlay on the whole book — full hedge-ratio sweep, real-time expanding & rolling-beta **dynamic hedging**, and stress-gated hybrids.

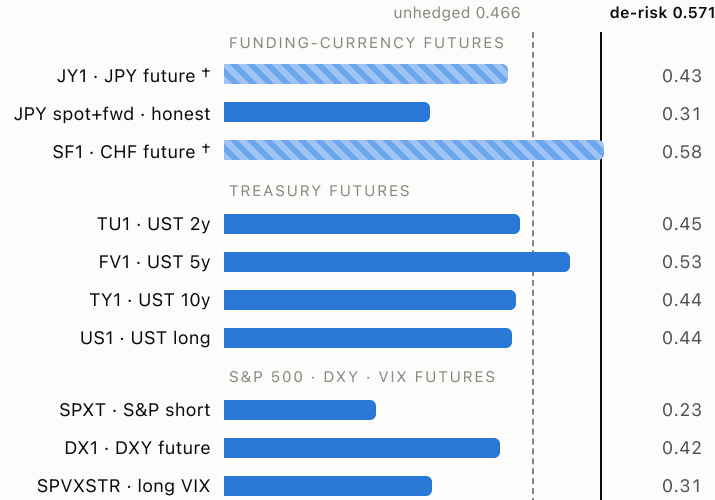
Part B — five futures classes, one yardstick. Every instrument (JY1, SF1, TU1, FV1, TY1, US1, SPXT/ES1, DX1, UX1/SPVXSTR) vs the book, with each hedged Sharpe **decomposed into variance reduction vs the instrument's own return** — so an instrument's bull market or a data artifact can't masquerade as hedging skill.

Data honesty. Real 2026-08 futures pulls (JY1, SF1, SPXT), gated to be static values; every generic futures series is cross-checked against a spot+forward replica to strip the **roll-splice artifact** (~200 bp/yr of phantom return in JY1/SF1; +46%/yr in VIX).

02 KEY FINDINGS

- **Nothing investable beats de-risking.** Judged against cutting exposure in stress (Sharpe 0.571), every honest futures overlay fails — most also land below the unhedged 0.466.
- **JPY is real insurance at full price.** Best tail payoff (+32–34 bp/day) and ~10% vol reduction, but the carry premium eats it; max drawdown gets *worse* (2013–15 JPY collapse).
- **Dynamic hedging doesn't rescue it** — the JPY beta never changes sign, so rolling re-estimation adds noise, not timing: dynamic 0.25–0.37 vs static 0.31–0.43.
- **The funding legs aren't the crash risk.** Cutting JPY/CHF legs in stress *hurts* (0.119 → 0.068–0.091 on the G10 book); the stress factor is broad risk appetite, spread across all legs.
- **Artifact watch.** The two rows that nominally clear the bar are both measurement artifacts: UX1 generic (phantom +46%/yr roll gain) and SF1 (0.576 → 0.469 on replica pricing).

03 STATIC VARIANCE-MINIMISING HEDGE, BY INSTRUMENT — NET SHARPE



+ generic-series roll-splice inflates these rows (~200 bp/yr of non-earnable return); SF1's nominal "beat" of the 0.571 bar drops to 0.469 on artifact-free replica pricing. Axis 0 → 0.65 Sharpe; each row on its clean joined sample.

04 VERDICT BY CLASS

INSTRUMENT CLASS	STATIC	DYNAMIC	WHAT IT'S ACTUALLY FOR
JPY / CHF futures	×	×	Honest tail insurance, fully priced — stress-gated & small, only under a drawdown mandate
Treasury futures	×	×	Free but weak stabiliser (vol −3–4%); best "no own-return" duration expression
S&P 500 futures	×	×	Wrong way — book is long-equity-beta; short sells the ERP (−2.7%/yr, t = −3.1). Trigger-only
DXY futures	×	×	Structurally blind: carry sort ≈ dollar-neutral, DXY has no EM. Third confirmed null
VIX futures	×	×	Most reliable crash correlation, steepest premium (−33%/yr roll) — tiny, episodic only

05 THE JPY OVERLAY, EVERY WAY IT WAS SIZED (A.3, HONEST REPLICA PRICING · IDENTICAL STRESS TRIGGER)

0.54

de-risk the book
(no hedge at all)

>

0.44

unhedged
book

>

0.39

stress-gated h*
(long JPY in stress only)

>

0.31

static h* = 0.52
(always-on hedge)

>

0.25–0.27

dynamic rolling / expanding β
(textbook dynamic hedging)

BOTTOM LINE

No futures overlay — static or dynamic — makes this book better. Everything that pays in a carry crash is priced as insurance and charges for it on every calm day; everything free is too weak or blind to the book's EM-risk-appetite exposure. The cheapest crash protection is to **condition the book itself**: de-risk on the FX-vol trigger (Sharpe 0.466 → 0.571), or the stress-triggered EM-equity short — not an always-on futures hedge.